

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df=pd.read_csv('/content/creditcard.csv')
```

1.DATA INSPECTION

df

|        | Time     | V1         | V2        | V3        | V4        | V5        | V6        | V7        | V8        | V9        | ... | V21       |
|--------|----------|------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----|-----------|
| 0      | 0.0      | -1.359807  | -0.072781 | 2.536347  | 1.378155  | -0.338321 | 0.462388  | 0.239599  | 0.098698  | 0.363787  | ... | -0.018307 |
| 1      | 0.0      | 1.191857   | 0.266151  | 0.166480  | 0.448154  | 0.060018  | -0.082361 | -0.078803 | 0.085102  | -0.255425 | ... | -0.225775 |
| 2      | 1.0      | -1.358354  | -1.340163 | 1.773209  | 0.379780  | -0.503198 | 1.800499  | 0.791461  | 0.247676  | -1.514654 | ... | 0.247998  |
| 3      | 1.0      | -0.966272  | -0.185226 | 1.792993  | -0.863291 | -0.010309 | 1.247203  | 0.237609  | 0.377436  | -1.387024 | ... | -0.108300 |
| 4      | 2.0      | -1.158233  | 0.877737  | 1.548718  | 0.403034  | -0.407193 | 0.095921  | 0.592941  | -0.270533 | 0.817739  | ... | -0.009431 |
| ...    | ...      | ...        | ...       | ...       | ...       | ...       | ...       | ...       | ...       | ...       | ... | ...       |
| 284802 | 172786.0 | -11.881118 | 10.071785 | -9.834783 | -2.066656 | -5.364473 | -2.606837 | -4.918215 | 7.305334  | 1.914428  | ... | 0.213454  |
| 284803 | 172787.0 | -0.732789  | -0.055080 | 2.035030  | -0.738589 | 0.868229  | 1.058415  | 0.024330  | 0.294869  | 0.584800  | ... | 0.214205  |
| 284804 | 172788.0 | 1.919565   | -0.301254 | -3.249640 | -0.557828 | 2.630515  | 3.031260  | -0.296827 | 0.708417  | 0.432454  | ... | 0.232045  |
| 284805 | 172788.0 | -0.240440  | 0.530483  | 0.702510  | 0.689799  | -0.377961 | 0.623708  | -0.686180 | 0.679145  | 0.392087  | ... | 0.265245  |
| 284806 | 172792.0 | -0.533413  | -0.189733 | 0.703337  | -0.506271 | -0.012546 | -0.649617 | 1.577006  | -0.414650 | 0.486180  | ... | 0.261057  |

284807 rows × 31 columns

df.head()

|   | Time | V1        | V2        | V3       | V4        | V5        | V6        | V7        | V8        | V9        | ... | V21       | V22       |
|---|------|-----------|-----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----|-----------|-----------|
| 0 | 0.0  | -1.359807 | -0.072781 | 2.536347 | 1.378155  | -0.338321 | 0.462388  | 0.239599  | 0.098698  | 0.363787  | ... | -0.018307 | 0.277838  |
| 1 | 0.0  | 1.191857  | 0.266151  | 0.166480 | 0.448154  | 0.060018  | -0.082361 | -0.078803 | 0.085102  | -0.255425 | ... | -0.225775 | -0.638672 |
| 2 | 1.0  | -1.358354 | -1.340163 | 1.773209 | 0.379780  | -0.503198 | 1.800499  | 0.791461  | 0.247676  | -1.514654 | ... | 0.247998  | 0.771679  |
| 3 | 1.0  | -0.966272 | -0.185226 | 1.792993 | -0.863291 | -0.010309 | 1.247203  | 0.237609  | 0.377436  | -1.387024 | ... | -0.108300 | 0.005274  |
| 4 | 2.0  | -1.158233 | 0.877737  | 1.548718 | 0.403034  | -0.407193 | 0.095921  | 0.592941  | -0.270533 | 0.817739  | ... | -0.009431 | 0.798278  |

5 rows × 31 columns

df.tail()

|        | Time     | V1         | V2        | V3        | V4        | V5        | V6        | V7        | V8        | V9       | ... | V21      |
|--------|----------|------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|-----|----------|
| 284802 | 172786.0 | -11.881118 | 10.071785 | -9.834783 | -2.066656 | -5.364473 | -2.606837 | -4.918215 | 7.305334  | 1.914428 | ... | 0.213454 |
| 284803 | 172787.0 | -0.732789  | -0.055080 | 2.035030  | -0.738589 | 0.868229  | 1.058415  | 0.024330  | 0.294869  | 0.584800 | ... | 0.214205 |
| 284804 | 172788.0 | 1.919565   | -0.301254 | -3.249640 | -0.557828 | 2.630515  | 3.031260  | -0.296827 | 0.708417  | 0.432454 | ... | 0.232045 |
| 284805 | 172788.0 | -0.240440  | 0.530483  | 0.702510  | 0.689799  | -0.377961 | 0.623708  | -0.686180 | 0.679145  | 0.392087 | ... | 0.265245 |
| 284806 | 172792.0 | -0.533413  | -0.189733 | 0.703337  | -0.506271 | -0.012546 | -0.649617 | 1.577006  | -0.414650 | 0.486180 | ... | 0.261057 |

5 rows × 31 columns

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 284807 entries, 0 to 284806
Data columns (total 31 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Time    284807 non-null   float64
1   V1      284807 non-null   float64
2   V2      284807 non-null   float64
```

```
3 V3 284807 non-null float64
4 V4 284807 non-null float64
5 V5 284807 non-null float64
6 V6 284807 non-null float64
7 V7 284807 non-null float64
8 V8 284807 non-null float64
9 V9 284807 non-null float64
10 V10 284807 non-null float64
11 V11 284807 non-null float64
12 V12 284807 non-null float64
13 V13 284807 non-null float64
14 V14 284807 non-null float64
15 V15 284807 non-null float64
16 V16 284807 non-null float64
17 V17 284807 non-null float64
18 V18 284807 non-null float64
19 V19 284807 non-null float64
20 V20 284807 non-null float64
21 V21 284807 non-null float64
22 V22 284807 non-null float64
23 V23 284807 non-null float64
24 V24 284807 non-null float64
25 V25 284807 non-null float64
26 V26 284807 non-null float64
27 V27 284807 non-null float64
28 V28 284807 non-null float64
29 Amount 284807 non-null float64
30 Class 284807 non-null int64
dtypes: float64(30), int64(1)
memory usage: 67.4 MB
```

```
df.describe()
```

|       | Time          | V1            | V2            | V3            | V4            | V5            | V6            | V7            |               |
|-------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| count | 284807.000000 | 2.848070e+05  | 2.848070e+05  | 2.848070e+05  | 2.848070e+05  | 2.848070e+05  | 2.848070e+05  | 2.848070e+05  | 2.848070e+05  |
| mean  | 94813.859575  | 1.168375e-15  | 3.416908e-16  | -1.379537e-15 | 2.074095e-15  | 9.604066e-16  | 1.487313e-15  | -5.556467e-16 | 1.168375e-15  |
| std   | 47488.145955  | 1.958696e+00  | 1.651309e+00  | 1.516255e+00  | 1.415869e+00  | 1.380247e+00  | 1.332271e+00  | 1.237094e+00  | 1.958696e+00  |
| min   | 0.000000      | -5.640751e+01 | -7.271573e+01 | -4.832559e+01 | -5.683171e+00 | -1.137433e+02 | -2.616051e+01 | -4.355724e+01 | -7.271573e+01 |
| 25%   | 54201.500000  | -9.203734e-01 | -5.985499e-01 | -8.903648e-01 | -8.486401e-01 | -6.915971e-01 | -7.682956e-01 | -5.540759e-01 | -2.037338e-01 |
| 50%   | 84692.000000  | 1.810880e-02  | 6.548556e-02  | 1.798463e-01  | -1.984653e-02 | -5.433583e-02 | -2.741871e-01 | 4.010308e-02  | 2.037338e-01  |
| 75%   | 139320.500000 | 1.315642e+00  | 8.037239e-01  | 1.027196e+00  | 7.433413e-01  | 6.119264e-01  | 3.985649e-01  | 5.704361e-01  | 3.066667e-01  |
| max   | 172792.000000 | 2.454930e+00  | 2.205773e+01  | 9.382558e+00  | 1.687534e+01  | 3.480167e+01  | 7.330163e+01  | 1.205895e+02  | 2.037338e-01  |

8 rows × 31 columns

SHAPE OF THE DATASET

```
df.shape
```

(284807, 31)

```
df.columns
```

```
Index(['Time', 'V1', 'V2', 'V3', 'V4', 'V5', 'V6', 'V7', 'V8', 'V9', 'V10',
      'V11', 'V12', 'V13', 'V14', 'V15', 'V16', 'V17', 'V18', 'V19', 'V20',
      'V21', 'V22', 'V23', 'V24', 'V25', 'V26', 'V27', 'V28', 'Amount',
      'Class'],
      dtype='object')
```

```
df.dtypes
```

|               | 0       |
|---------------|---------|
| <b>Time</b>   | float64 |
| <b>V1</b>     | float64 |
| <b>V2</b>     | float64 |
| <b>V3</b>     | float64 |
| <b>V4</b>     | float64 |
| <b>V5</b>     | float64 |
| <b>V6</b>     | float64 |
| <b>V7</b>     | float64 |
| <b>V8</b>     | float64 |
| <b>V9</b>     | float64 |
| <b>V10</b>    | float64 |
| <b>V11</b>    | float64 |
| <b>V12</b>    | float64 |
| <b>V13</b>    | float64 |
| <b>V14</b>    | float64 |
| <b>V15</b>    | float64 |
| <b>V16</b>    | float64 |
| <b>V17</b>    | float64 |
| <b>V18</b>    | float64 |
| <b>V19</b>    | float64 |
| <b>V20</b>    | float64 |
| <b>V21</b>    | float64 |
| <b>V22</b>    | float64 |
| <b>V23</b>    | float64 |
| <b>V24</b>    | float64 |
| <b>V25</b>    | float64 |
| <b>V26</b>    | float64 |
| <b>V27</b>    | float64 |
| <b>V28</b>    | float64 |
| <b>Amount</b> | float64 |
| <b>Class</b>  | int64   |

dtype: object

## CHECK MISSING VALUES

```
df.isnull().sum()
```

|               | $\emptyset$ |
|---------------|-------------|
| <b>Time</b>   | 0           |
| <b>V1</b>     | 0           |
| <b>V2</b>     | 0           |
| <b>V3</b>     | 0           |
| <b>V4</b>     | 0           |
| <b>V5</b>     | 0           |
| <b>V6</b>     | 0           |
| <b>V7</b>     | 0           |
| <b>V8</b>     | 0           |
| <b>V9</b>     | 0           |
| <b>V10</b>    | 0           |
| <b>V11</b>    | 0           |
| <b>V12</b>    | 0           |
| <b>V13</b>    | 0           |
| <b>V14</b>    | 0           |
| <b>V15</b>    | 0           |
| <b>V16</b>    | 0           |
| <b>V17</b>    | 0           |
| <b>V18</b>    | 0           |
| <b>V19</b>    | 0           |
| <b>V20</b>    | 0           |
| <b>V21</b>    | 0           |
| <b>V22</b>    | 0           |
| <b>V23</b>    | 0           |
| <b>V24</b>    | 0           |
| <b>V25</b>    | 0           |
| <b>V26</b>    | 0           |
| <b>V27</b>    | 0           |
| <b>V28</b>    | 0           |
| <b>Amount</b> | 0           |
| <b>Class</b>  | 0           |

**dtype:** int64

```
df.isnull().mean()*100
```

|               | $\emptyset$ |
|---------------|-------------|
| <b>Time</b>   | 0.0         |
| <b>V1</b>     | 0.0         |
| <b>V2</b>     | 0.0         |
| <b>V3</b>     | 0.0         |
| <b>V4</b>     | 0.0         |
| <b>V5</b>     | 0.0         |
| <b>V6</b>     | 0.0         |
| <b>V7</b>     | 0.0         |
| <b>V8</b>     | 0.0         |
| <b>V9</b>     | 0.0         |
| <b>V10</b>    | 0.0         |
| <b>V11</b>    | 0.0         |
| <b>V12</b>    | 0.0         |
| <b>V13</b>    | 0.0         |
| <b>V14</b>    | 0.0         |
| <b>V15</b>    | 0.0         |
| <b>V16</b>    | 0.0         |
| <b>V17</b>    | 0.0         |
| <b>V18</b>    | 0.0         |
| <b>V19</b>    | 0.0         |
| <b>V20</b>    | 0.0         |
| <b>V21</b>    | 0.0         |
| <b>V22</b>    | 0.0         |
| <b>V23</b>    | 0.0         |
| <b>V24</b>    | 0.0         |
| <b>V25</b>    | 0.0         |
| <b>V26</b>    | 0.0         |
| <b>V27</b>    | 0.0         |
| <b>V28</b>    | 0.0         |
| <b>Amount</b> | 0.0         |
| <b>Class</b>  | 0.0         |

**dtype:** float64

```
df.notnull().sum()
```

```
0
Time 284807
V1 284807
V2 284807
V3 284807
V4 284807
V5 284807
V6 284807
V7 284807
V8 284807
V9 284807
V10 284807
V11 284807
V12 284807
V13 284807
V14 284807
V15 284807
V16 284807
V17 284807
V18 284807
V19 284807
V20 284807
V21 284807
V22 284807
V23 284807
V24 284807
V25 284807
V26 284807
V27 284807
V28 284807
Amount 284807
Class 284807
```

```
dtype: int64
```

```
df.nunique()
```

| 0      |        |
|--------|--------|
| Time   | 124592 |
| V1     | 275663 |
| V2     | 275663 |
| V3     | 275663 |
| V4     | 275663 |
| V5     | 275663 |
| V6     | 275663 |
| V7     | 275663 |
| V8     | 275663 |
| V9     | 275663 |
| V10    | 275663 |
| V11    | 275663 |
| V12    | 275663 |
| V13    | 275663 |
| V14    | 275663 |
| V15    | 275663 |
| V16    | 275663 |
| V17    | 275663 |
| V18    | 275663 |
| V19    | 275663 |
| V20    | 275663 |
| V21    | 275663 |
| V22    | 275663 |
| V23    | 275663 |
| V24    | 275663 |
| V25    | 275663 |
| V26    | 275663 |
| V27    | 275663 |
| V28    | 275663 |
| Amount | 32767  |
| Class  | 2      |

dtype: int64

df.index

RangeIndex(start=0, stop=284807, step=1)

df.memory\_usage()

| 0      |         |
|--------|---------|
| Index  | 132     |
| Time   | 2278456 |
| V1     | 2278456 |
| V2     | 2278456 |
| V3     | 2278456 |
| V4     | 2278456 |
| V5     | 2278456 |
| V6     | 2278456 |
| V7     | 2278456 |
| V8     | 2278456 |
| V9     | 2278456 |
| V10    | 2278456 |
| V11    | 2278456 |
| V12    | 2278456 |
| V13    | 2278456 |
| V14    | 2278456 |
| V15    | 2278456 |
| V16    | 2278456 |
| V17    | 2278456 |
| V18    | 2278456 |
| V19    | 2278456 |
| V20    | 2278456 |
| V21    | 2278456 |
| V22    | 2278456 |
| V23    | 2278456 |
| V24    | 2278456 |
| V25    | 2278456 |
| V26    | 2278456 |
| V27    | 2278456 |
| V28    | 2278456 |
| Amount | 2278456 |
| Class  | 2278456 |

dtype: int64

df.size

8829017

df.ndim

2

df.count(axis='columns')

```

      0
0    31
1    31
2    31
3    31
4    31
...   ...
284802 31
284803 31
284804 31
284805 31
284806 31
284807 rows x 1 columns

```

**dtype:** int64

### CHECK DUPLICATES

```
df.duplicated().sum()
```

```
np.int64(1081)
```

```
df = df.drop_duplicates()
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

### INSPECT TARGET VARIABLE

```
df['Class'].value_counts()
```

```

      count
Class
0    283253
1      473

```

**dtype:** int64

```
df['Class'].value_counts(normalize=True)*100
```

```

      proportion
Class
0    99.83329
1     0.16671

```

**dtype:** float64

### SKEWNESS

0 ( $\approx 0$ ) Symmetric distribution

Positive ( $>0$ )  $\rightarrow$  positive skew

Negative ( $<0$ )  $\rightarrow$  negative skew

```
df.skew()
```

|               | $\theta$  |
|---------------|-----------|
| <b>Time</b>   | -0.035581 |
| <b>V1</b>     | -3.273271 |
| <b>V2</b>     | -4.695162 |
| <b>V3</b>     | -2.151984 |
| <b>V4</b>     | 0.671504  |
| <b>V5</b>     | -2.414079 |
| <b>V6</b>     | 1.829880  |
| <b>V7</b>     | 2.890271  |
| <b>V8</b>     | -8.310970 |
| <b>V9</b>     | 0.537663  |
| <b>V10</b>    | 1.252967  |
| <b>V11</b>    | 0.344074  |
| <b>V12</b>    | -2.199008 |
| <b>V13</b>    | 0.064293  |
| <b>V14</b>    | -1.918804 |
| <b>V15</b>    | -0.309659 |
| <b>V16</b>    | -1.051161 |
| <b>V17</b>    | -3.690497 |
| <b>V18</b>    | -0.248661 |
| <b>V19</b>    | 0.108312  |
| <b>V20</b>    | -2.043121 |
| <b>V21</b>    | 2.820033  |
| <b>V22</b>    | -0.182330 |
| <b>V23</b>    | -5.867221 |
| <b>V24</b>    | -0.552129 |
| <b>V25</b>    | -0.415744 |
| <b>V26</b>    | 0.580292  |
| <b>V27</b>    | -0.753804 |
| <b>V28</b>    | 11.555115 |
| <b>Amount</b> | 16.978803 |
| <b>Class</b>  | 24.430545 |

dtype: float64

### CORRELATION

+1 → perfect positive correlation (both increase together)

-1 → perfect negative correlation (one increases, other decreases)

0 → no linear correlation

```
corr=df.corr()
```

## 2.DATA CLEANING

```
df=df.dropna()
```

```
df.isnull().sum()
```

|               | $\emptyset$ |
|---------------|-------------|
| <b>Time</b>   | 0           |
| <b>V1</b>     | 0           |
| <b>V2</b>     | 0           |
| <b>V3</b>     | 0           |
| <b>V4</b>     | 0           |
| <b>V5</b>     | 0           |
| <b>V6</b>     | 0           |
| <b>V7</b>     | 0           |
| <b>V8</b>     | 0           |
| <b>V9</b>     | 0           |
| <b>V10</b>    | 0           |
| <b>V11</b>    | 0           |
| <b>V12</b>    | 0           |
| <b>V13</b>    | 0           |
| <b>V14</b>    | 0           |
| <b>V15</b>    | 0           |
| <b>V16</b>    | 0           |
| <b>V17</b>    | 0           |
| <b>V18</b>    | 0           |
| <b>V19</b>    | 0           |
| <b>V20</b>    | 0           |
| <b>V21</b>    | 0           |
| <b>V22</b>    | 0           |
| <b>V23</b>    | 0           |
| <b>V24</b>    | 0           |
| <b>V25</b>    | 0           |
| <b>V26</b>    | 0           |
| <b>V27</b>    | 0           |
| <b>V28</b>    | 0           |
| <b>Amount</b> | 0           |
| <b>Class</b>  | 0           |

**dtype:** int64

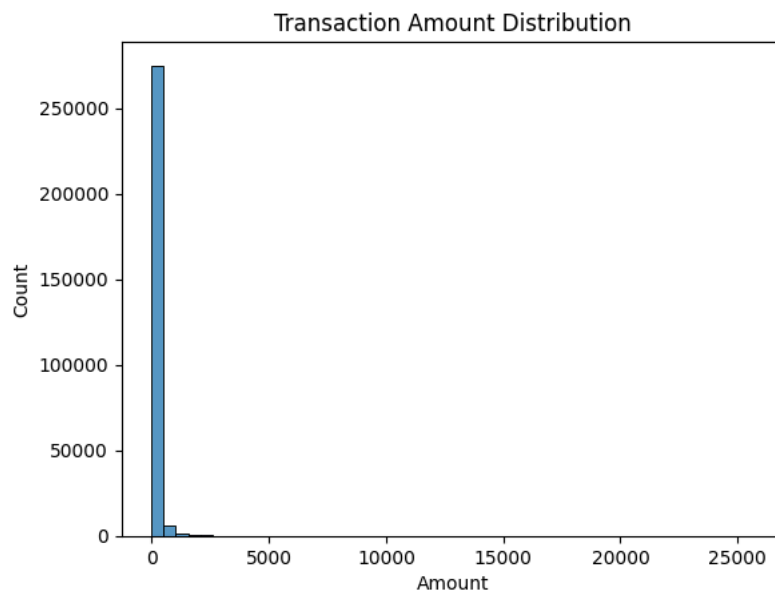
```
df=df.drop_duplicates()
```

## EDA-EXPLORATORY DATA ANALYSIS

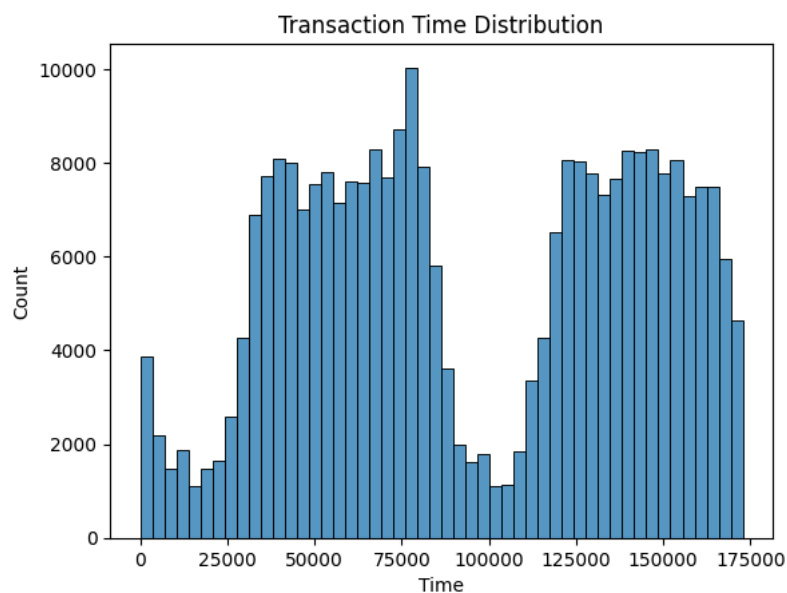
```
plt.figure(figsize=(6,4))
sns.countplot(x='Class',data=df)
plt.title('Fraud vs Non-Fraud detection')
plt.show()
```



```
sns.histplot(df['Amount'], bins=50)
plt.title("Transaction Amount Distribution")
plt.show()
```

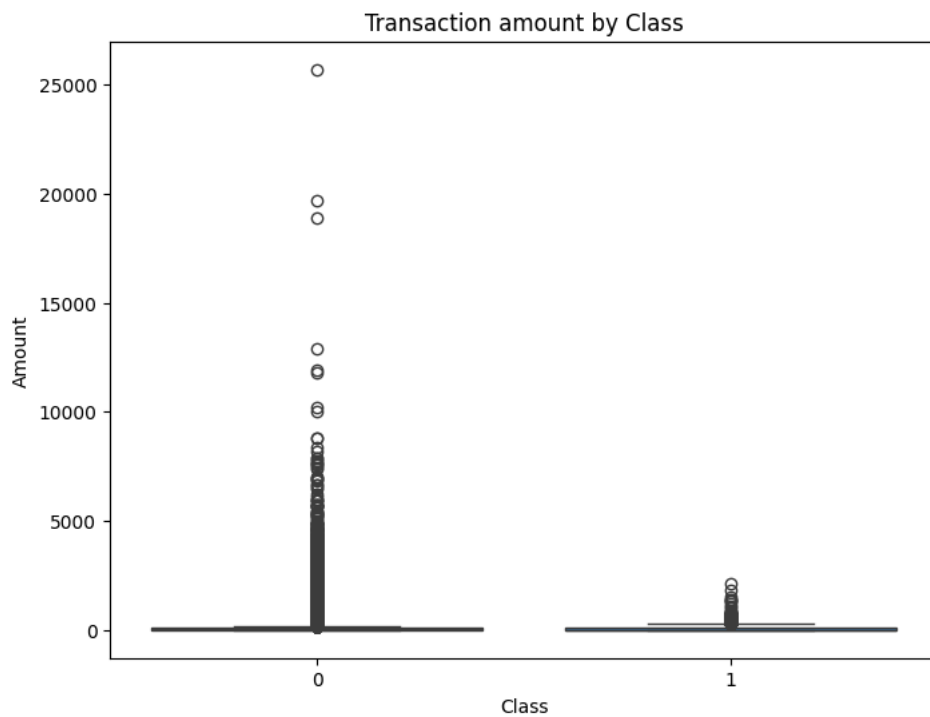


```
sns.histplot(df['Time'], bins=50)
plt.title("Transaction Time Distribution")
plt.show()
```

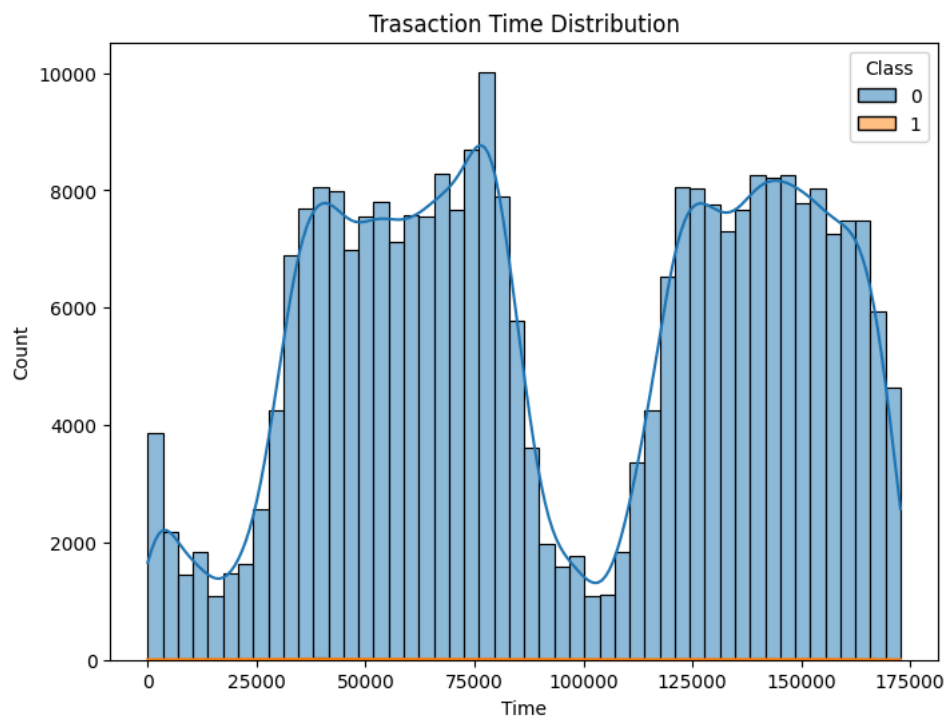


```
plt.figure(figsize=(8,6))
sns.boxplot(x='Class',y='Amount',data=df)
plt.title('Transaction amount by Class')
plt.show()

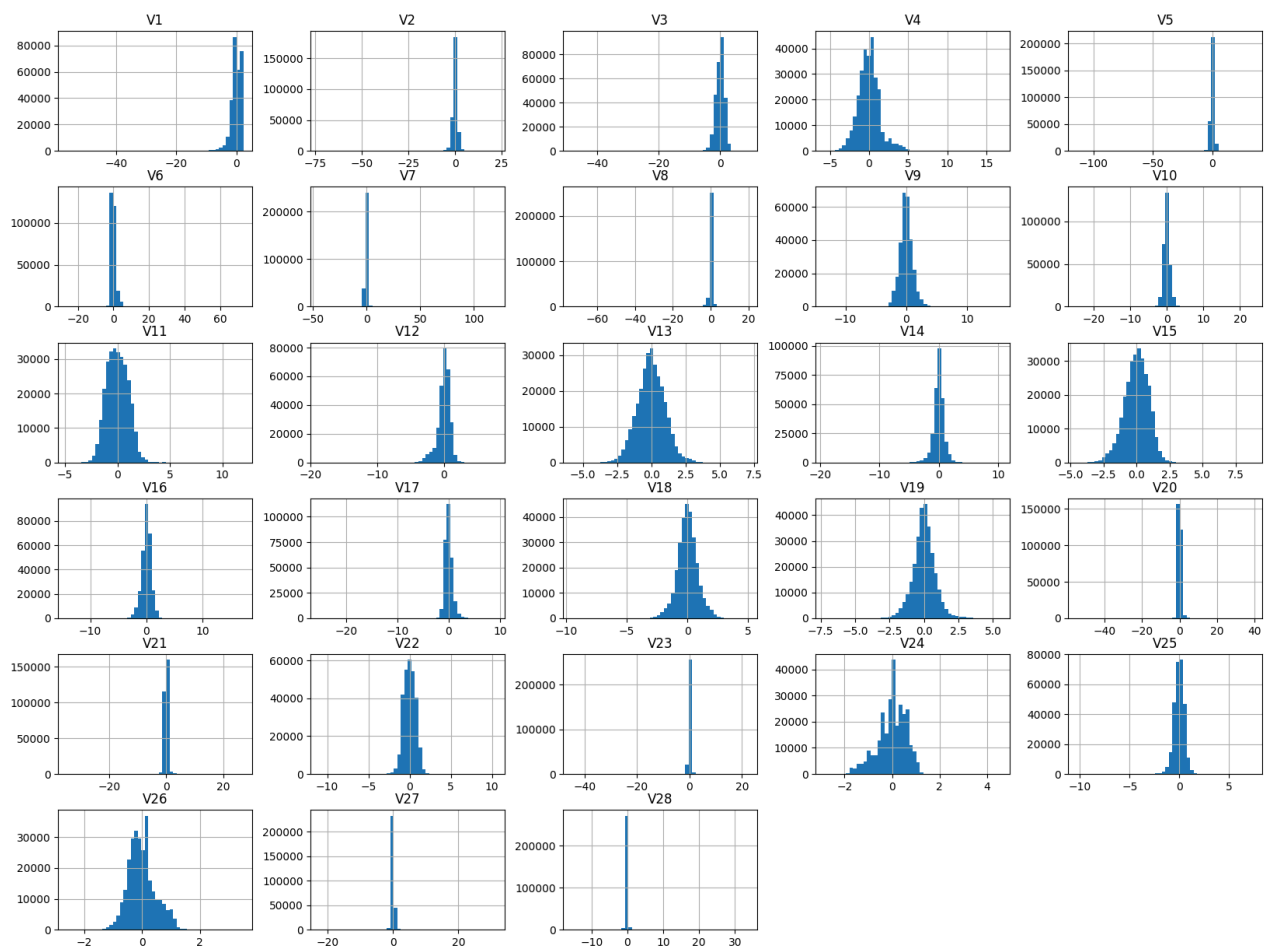
)
```



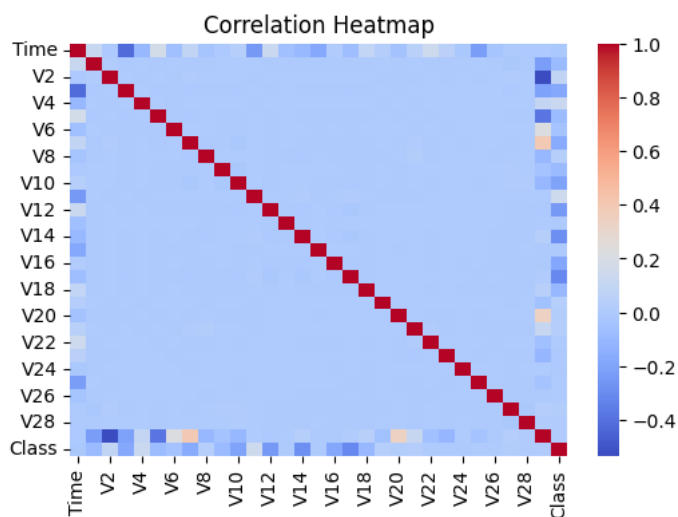
```
plt.figure(figsize=(8,6))
sns.histplot(data=df,x='Time',kde=True,hue='Class',bins=50)
plt.title("Trasaction Time Distribution")
plt.show()
```



```
df.iloc[:,1:29].hist(figsize=(20,15), bins=50)
plt.show()
```

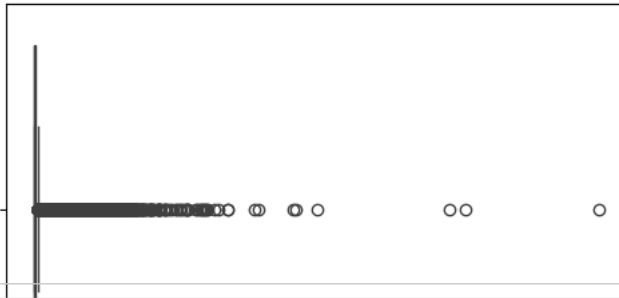


```
plt.figure(figsize=(6,4))
sns.heatmap(corr, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



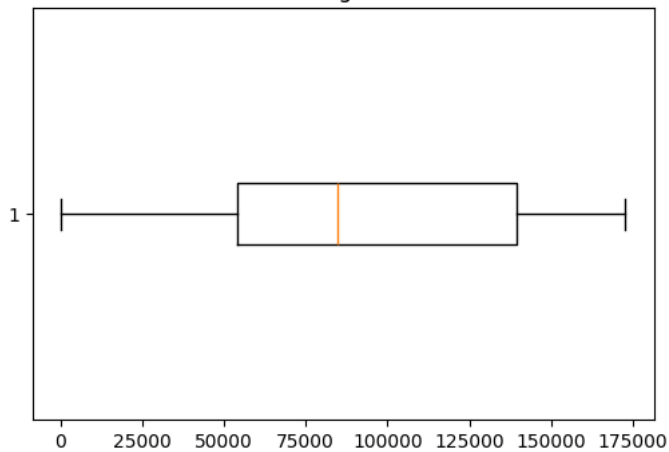
```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['Amount'])
plt.title("Amount Outliers")
plt.show()
```

Amount Outliers



```
def box_out(df):  
    plt.figure(figsize=(6,4))  
    plt.boxplot(df, vert=False)  
    plt.title("Detecting outliers")  
    plt.show()  
box_out(df['Time'])
```

Detecting outliers



```
plt.figure(figsize=(6,4))  
sns.scatterplot(x='Time', y='Amount', hue='Class', data=df, alpha=0.5)  
plt.title("Transaction Time vs Amount")  
plt.show()
```

Transaction Time vs Amount

